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To: Powertrain Dynamics Group
From: Edgar Meyhofer and Pramod Reddy,
Directors, Wolverine System Dynamics
Date: Sept 4th, 2026
Subject: Drive Shaft Vibrations in the Gulo 3000

During the last regular planning meeting with Mesa Hybrid Vehicles Inc., a major vibration problem was identified in its new prototype hybrid electric model, the *Gulo 3000*. The design team believes that this vibration is associated with the drive shaft connecting the DC motor to the front-wheels of the car. The vibration manifests itself transiently when the DC motor is turned on/off and was found to be sufficiently large to be uncomfortable to the driver and passengers. The general conclusion from our meeting was that the magnitude of the vibration was unacceptable and needs to be reduced by at least 30%.

From discussions with team members of Mesa's *Gulo* design group, it appears that the "flexibility" of the drive-axle, which was assumed to be rigid in their design work, is the cause of the undesirable vibrations. Fortunately, we have available in-house, from previous work, an appropriate physical model of the motor and drive-axle system that has been instrumented and is accessible for your use in the Systems Lab. Your group is assigned the task of first developing a motor/drive-axle model to better understand the system and to subsequently remedy the vibration problem.

Specifically, you need to

- (1) validate that the vibration is indeed due to shaft flexibility,
- (2) characterize the system and develop a mathematical model, and
- (3) perform simulations using the validated model to identify possible design changes to solve the vibration problem.

Mesa would like to review several possible solutions to the vibration problem and select, based on a broad range of criteria beyond the actual drive shaft design, the best design for the final powertrain. Therefore, you are charged with identifying the four most effective solutions to achieve a 30% reduction in the vibrations.

This project has an extremely short timeline and must have your highest priority. You will be presenting the results of your experimental work and analysis in a poster session in the week of September 21. Because of the significance of this vibration problem, the poster session will be attended by Mesa engineers and executives with diverse technical backgrounds. Please ensure that your poster uses clear visual elements to provide a brief, broadly comprehensible introduction and conclusions alongside the central technical data. In addition, a short, three-page handout summarizing the key findings from your work is due to us in the same week during your poster presentation.